

Gauge: Distribution-Free Conformal Coverage for IVIM Parameter Maps, and the Identifiability Wall in the Pseudo-Diffusion Compartment

Avery Karlin

ORCID: [0000-0003-3848-6782](https://orcid.org/0000-0003-3848-6782)

Abstract

Purpose: To bring distribution-free conformal prediction to intravoxel incoherent motion (IVIM) parameter mapping, benchmark it against model-based uncertainty quantification (UQ), and test whether its advantage concentrates in the ill-posed pseudo-diffusion compartment D^* .

Methods: On a controlled, seeded synthetic cohort built from our own bi-exponential forward model with Rician noise, we compare split-conformal and conformalized quantile regression (CQR) against four model-based baselines (probabilistic neural network, mixture-density deep ensemble, deep ensemble, per-voxel Bayesian MCMC), measuring marginal and ($\text{SNR} \times \text{true-}D^*$ -tercile) conditional coverage and sharpness, with empirical [5, 95] confidence bands on the conditional quantities from a 16-seed re-run. A Fisher-information / Cramér–Rao analysis diagnoses the residual gap; weighted (nonexchangeable) conformal and a label-free deployment monitor stress exchangeability; a qualitative in-vivo demonstration is included.

Results: Model-based IVIM UQ is overconfident *broadly* across D , D^* , and f (not specifically pseudo-diffusion), so the naive marginal hypothesis is rejected. Conformal and conformalized methods restore near-nominal marginal coverage everywhere ($|\text{gap}| \leq 0.024$), and conformalizing the MDN is the sharpest valid recipe (0.65–0.77 $\times$ pure-CQR width). The real failure is conditional: the high- D^* regime under-covers for every method (robustly at low-to-moderate SNR), and no label-free method closes the gap because the Cramér–Rao bound on D^* reaches the regime-bin width – an irreducible identifiability limit. The wall is robust to the b-value scheme; weighted conformal recovers covariate shift but not forward-model misspecification; the monitor fires before coverage fails but is blind to the latent gap.

Conclusion: Conformal prediction supplies the finite-sample coverage guarantee model-based IVIM UQ lacks, and conformalizing a strong base is sharpest. The high- D^* compartment is an identifiability limit we characterize, not solve.

Keywords: intravoxel incoherent motion (IVIM); uncertainty quantification; conformal prediction; identifiability; pseudo-diffusion; quantitative MRI.

1 Introduction

IVIM imaging [1] separates microcirculatory perfusion from tissue diffusion by fitting the bi-exponential signal

$$\frac{S(b)}{S_0} = f e^{-bD^*} + (1 - f) e^{-bD}, \quad (1)$$

where b is the diffusion weighting (s/mm^2), D the tissue diffusion coefficient, $D^* \gg D$ the pseudo-diffusion coefficient, and f the perfusion fraction. Equation (1) is notoriously ill-posed: D^* is poorly constrained because the fast-decaying perfusion term contributes only at low b , and small perfusion

fractions make it nearly unidentifiable. Clinical use of IVIM parameter maps therefore demands trustworthy *per-voxel uncertainty*, not just point estimates.

The dominant approach to IVIM UQ is *model-based*: probabilistic neural networks, deep ensembles, mixture-density networks, and Bayesian fitting place a distribution over parameters and read intervals from it. These methods make distributional assumptions and carry *no finite-sample coverage guarantee*; the most comprehensive IVIM-UQ study to date [2] documents that such models are well-calibrated for D and f but *overconfident in D^** . *Conformal prediction* [3–5] offers the complementary property: under exchangeability of calibration and test data, it produces intervals with finite-sample, distribution-free marginal coverage for *any* base predictor – a biased base only widens the interval, it does not break coverage. Conformalized quantile regression (CQR) [6] makes these intervals input-adaptive. Conformal UQ has been applied to quantitative MRI relaxometry and field mapping [7], but never to the ill-posed IVIM inverse problem, and never benchmarked head-to-head against model-based IVIM UQ.

This paper closes that gap and, in doing so, *reframes* the question. We began with the hypothesis that conformal prediction’s advantage would concentrate in the unstable D^* (and secondarily f) compartment, exactly where model-based UQ is documented weak. The data reject the *marginal* form of that hypothesis and replace it with a sharper, IVIM-specific finding. Our contributions are:

1. **Conformal/CQR coverage for IVIM** with validated finite-sample marginal coverage for (D, D^*, f) (Sec. 2).
2. **A conformal-vs-model-based benchmark** showing model-based IVIM UQ is overconfident *broadly*; conformal restores coverage in every compartment; and conformalizing the MDN is the sharpest valid recipe (Sec. 3.1).
3. **The conditional finding and its diagnosis**: the high- D^* regime under-covers universally, no label-free method closes the gap, and a Cramér–Rao analysis shows this is an *irreducible identifiability limit* (Secs. 3.2–3.3).
4. **Robustness, acquisition-sensitivity, and an in-vivo demonstration**: weighted conformal recovers covariate shift but not misspecification, an observable monitor flags deployment-validity breaks before coverage fails, the wall is robust to the b-value scheme, and the in-vivo demo is honestly qualitative (Secs. 3.4–3.6).

Every number in this paper traces to a gated, seeded checkpoint output; a machine-checked consistency report is included (Sec. 6).

2 Methods

Forward model and synthetic cohort. Conformal calibration requires *labeled* data, which in-vivo IVIM cannot provide, so the cohort is synthetic and built from our own implementation of Eq. (1) with Rician noise ($\sigma = S_0/\text{SNR}$). Parameters are drawn uniformly from published abdominal ranges: $D \in [0.5, 3.0] \times 10^{-3}$, $D^* \in [10, 100] \times 10^{-3} \text{ mm}^2/\text{s}$, $f \in [0.05, 0.40]$; SNR is drawn from $\{10, 20, 30, 50, 100\}$; the b-value scheme has 22 values $(0, 10, \dots, 100, 120, \dots, 200, 300, \dots, 800 \text{ s/mm}^2)$, dense at low b to separate D^* . The three splits (train/calibration/test = 4000/2000/3000) are i.i.d. draws from one seed (20260613), making calibration and test exchangeable. On clean (noise-free) signals the generator and the non-linear least-squares (NLLS) estimator recover the truth to a maximum relative error of 2.07×10^{-7} , confirming label self-consistency.

Conformal intervals. For a base point predictor $\hat{\theta}$, *split conformal* uses calibration nonconformity scores $s_i = |\theta_i - \hat{\theta}_i|$ and the radius $q = \lceil (n+1)(1-\alpha) \rceil$ -th smallest score, giving $[\hat{\theta} - q, \hat{\theta} + q]$ with marginal coverage in $[1 - \alpha, 1 - \alpha + 1/(n+1)]$. *CQR* [6] conformalizes a gradient-boosted quantile-regression base. We also *conformalize model-based* bases by treating their predictive quantiles as the CQR band, which restores the guarantee while keeping the model’s sharpness.

Model-based baselines. Four standard, independent model-based UQ baselines emit per-parameter intervals on the matched cohort: a single probabilistic NN (Gaussian NLL head); a deep ensemble of mixture-density networks [8, 9] following the Casali approach [2] with an aleatoric/epistemic split; a plain deep ensemble of point regressors (epistemic-only; deliberately weak); and a per-voxel Bayesian fit by component-wise adaptive Metropolis (fed the true noise level, so it is evaluated at its best). A parallel line of work uses normalizing-flow / simulation-based posterior estimation for microstructure inference; Gauge is deliberately *independent* of it (own forward model, own cohort, standard baselines) and does not use it as a baseline.

Conditional coverage and the identifiability diagnostic. We measure coverage on a (true- D^* tercile $\times$ SNR) grid – the terciles are a diagnostic only; the true D^* is unknown at test time. To ask whether a residual gap is a method failure or an information limit, we compute the Fisher information for $\theta = (D, D^*, f, S_0)$ under the high-SNR Gaussian approximation, $\mathcal{I} = J^\top J / \sigma^2$ with J the signal Jacobian, and the Cramér–Rao lower bound (CRLB) [10] $\text{CRLB}(\theta_k) = \sqrt{[\mathcal{I}^{-1}]_{kk}}$.

Robustness tooling. To stress exchangeability we use *weighted / nonexchangeable conformal* [11, 12], with importance weights $w(x) = dP_{\text{test}}/dP_{\text{cal}}$ estimated by a domain classifier on observable features, and *localized* [13] and *conditional* [14] conformal for the label-free conditioning attack. We also build a label-free *deployment-validity monitor* with two detector families (a Mahalanobis distance on observable signal-summary features and a residual-conformal test on NLLS fit residuals) that fires when an aggregate drift score exceeds a calibration-derived null threshold.

3 Results

3.1 Marginal benchmark: model-based UQ is broadly overconfident

Plain split conformal and CQR track nominal coverage within 0.03 for all parameters and all $\alpha \in \{0.05, 0.10, 0.20, 0.30\}$ on the held-out test split (e.g. split-conformal D^* at $\alpha=0.10$: realized 0.897 vs nominal 0.90), confirming the conformal layer is correct. Table 1 gives the head-to-head at $\alpha = 0.10$.

Averaging the gap over α , the marginal under-coverage is broad, not D^*/f -concentrated: per-method worst compartments are f (PNN, 0.125), f (MDN, 0.037), D^* (DeepEnsemble-Point, 0.452), and f (Bayesian, 0.045). 0/4 **methods** show D^*/f -concentrated under-coverage; for the MDN ensemble D^* is in fact among the *better*-covered parameters marginally (gap 0.029), because its large aleatoric variance widens its band. The naive marginal hypothesis is therefore *rejected*. Conformal and conformalized variants restore near-nominal coverage everywhere, with maximum conformalized gap 0.024 (Fig. 1). On sharpness (Fig. 2), the cost of restoring coverage is cheap for strong bases (MDN 1.05–1.07 $\times$, Bayesian 1.02–1.05 $\times$ the raw width) and ruinous for the overconfident weak base (DeepEnsemble-Point 3.23–3.76 $\times$). Crucially, the model’s band helps: *conformalized-MDN* is 0.73/0.77/0.65 $\times$ the width of pure CQR for $D/D^*/f$ at equal guaranteed coverage – the sharpest valid recipe.

Table 1: Marginal coverage on test at $\alpha = 0.10$ (nominal 0.90); gap = nominal – realized (positive = overconfident). Model-based methods under-cover broadly; conformal and conformalized variants restore coverage in every compartment. Marginal coverage is finite-sample *guaranteed*; values are the seed-0 (multi-seed index-0) run, with 16-seed MC-precision bands available in the supplementary CI table. From `results/benchmark_report.txt`.

arm	D cov (gap)	D^* cov (gap)	f cov (gap)
raw: PNN-Gaussian	0.787 (+0.113)	0.833 (+0.067)	0.767 (+0.133)
raw: MDN-DeepEnsemble	0.870 (+0.030)	0.876 (+0.024)	0.859 (+0.041)
raw: DeepEnsemble-Point	0.436 (+0.464)	0.425 (+0.475)	0.475 (+0.425)
raw: Bayesian-MCMC	0.876 (+0.024)	0.870 (+0.030)	0.855 (+0.045)
conformal: split-NLLS	0.899 (+0.001)	0.897 (+0.003)	0.907 (−0.007)
conformal: CQR-HGB	0.901 (−0.001)	0.902 (−0.002)	0.893 (+0.007)
conformalized: MDN	0.890 (+0.010)	0.901 (−0.001)	0.882 (+0.018)
conformalized: Bayesian	0.895 (+0.005)	0.894 (+0.006)	0.876 (+0.024)

Model-based UQ under-covers (below diagonal); conformal sits on the diagonal

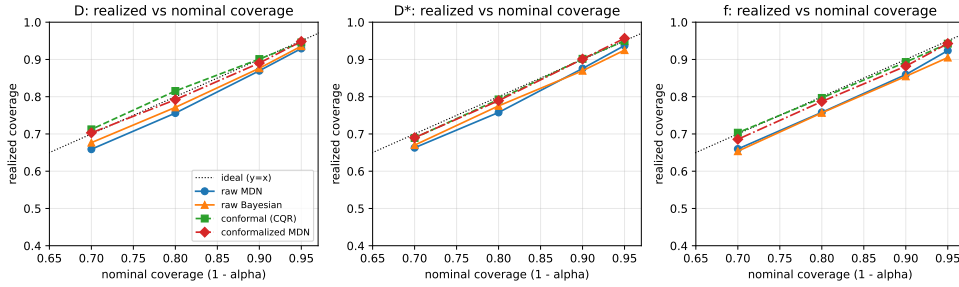


Figure 1: Realized vs nominal coverage per parameter. Raw model-based methods sit below the diagonal (overconfident); conformal and conformalized methods sit on it.

3.2 The conditional finding: high- D^* under-covers universally

Marginal coverage hides a compartment-specific failure (Fig. 3). On the ($\text{SNR} \times D^*$ -tercile) grid, the high- D^* tercile under-covers for *every* method at *every* SNR, while the mid- D^* tercile over-covers (≈ 0.99). Table 2 gives the high- D^* slice: even the best method (conformalized-MDN) reaches only 0.868[0.856, 0.889] marginal-in-regime and 0.786[0.751, 0.826] in its worst-SNR cell (16-seed mean and [5, 95] band). Critically, *Mondrian stratification by the known SNR does not fix it* – the failure axis is the *unknown* true D^* , not SNR. This is the refined, IVIM-specific form of the original hypothesis.

3.3 The high- D^* gap is an irreducible identifiability limit

We attacked the high- D^* gap with eleven *label-free* conditioning methods – plug-in Mondrian by estimated $\hat{D}^*$, localized conformal on observable features [13], and conditional conformal over an observable basis [14] – none of which uses the true D^* (Fig. 4). *None materially closes the gap*. The best label-free method (MDN with localized conformal) reaches 0.874[0.860, 0.899] marginal / 0.807[0.774, 0.845] worst-SNR, within sampling noise of the 0.786 baseline. The reason is structural, not methodological:

- **Routing error.** A plug-in $\hat{D}^*$ misroutes $\sim 33\%$ ([30, 35]% over 16 seeds) of true high- D^*

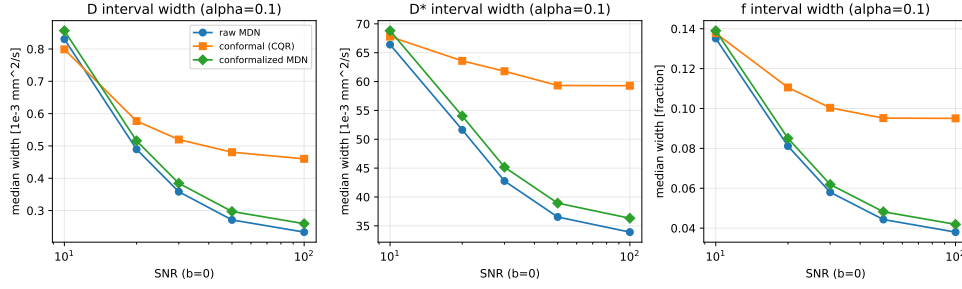


Figure 2: Median interval width vs SNR. Conformalized-MDN is the sharpest method with a guaranteed coverage level.

Table 2: High- D^* tercile coverage (D^* parameter, nominal 0.90). Every method under-covers; SNR-Mondrian does not close it. Across-seed means with 16-seed [5, 95] bands. From `results/multiseed_report.txt`.

arm	hi- D^* marginal	hi- D^* worst-SNR cell
raw-MDN	0.841 [0.829, 0.857]	0.753 [0.719, 0.789]
CQR (plain)	0.826 [0.809, 0.851]	0.761 [0.729, 0.795]
split (Mondrian/SNR)	0.829 [0.813, 0.844]	0.781 [0.743, 0.810]
CQR (Mondrian/SNR)	0.829 [0.811, 0.854]	0.789 [0.752, 0.828]
conformalized-MDN	0.868 [0.856, 0.889]	0.786 [0.751, 0.826]

voxels to a lower- D^* (smaller-radius) stratum: the estimate is least reliable exactly where it must route correctly. Mondrian-by- $\hat{D}^*$ attains nominal coverage *conditional on the observed stratum* ([0.90, 0.89, 0.90]) yet craters on the *true* high- D^* tercile (0.78 [0.76, 0.80]) – it controls the observable, not the latent axis.

- **The CRLB wall.** The CRLB on D^* grows $\sim 6\times$ from low to high D^* (an analytic, seed-invariant bound); the ratio of median CRLB(D^*) to the tercile bin width rises $0.34 \rightarrow 0.67 \rightarrow 1.10$ across terciles, the high tercile band [1.02, 1.15] (Fig. 5). In the high- D^* regime the CRLB *exceeds* the bin width: a voxel’s D^* cannot be localized to its own tercile from the data, so no observable can route it to a wider-interval stratum. Conformal interval width correctly tracks the per-voxel CRLB (log-log $r = 0.76$ [0.72, 0.77]): the intervals are honest, not broken.

Verdict. High- D^* regime-conditional coverage in IVIM is an *irreducible identifiability limit*. We therefore *characterize* the limit; we do not claim to solve it.

3.4 Robustness under deliberate exchangeability breaks

We deploy one conformal predictor and one monitor, calibrated on the i.i.d. Rician cohort, against four deliberate breaks (Table 3, Fig. 6). Every break miscalibrates coverage; the dangerous under-coverage reaches $D^* = 0.508$ [0.479, 0.543] under a harder-tissue prior shift and the worst parameter 0.592 [0.557, 0.637] under a low-SNR covariate shift. *Weighted / nonexchangeable conformal* [11, 12] recovers the covariate shifts (SNR worst-parameter 0.592 $\rightarrow$ 0.882, all within 0.018 of nominal; prior $D^* 0.508 \rightarrow 0.991$ [0.974, 1.000]) and is honestly limited on the tri-exponential forward-model

Conditional coverage of D^* : the hi- D^* row under-covers across all SNR (and resists SNR-Mondrian)

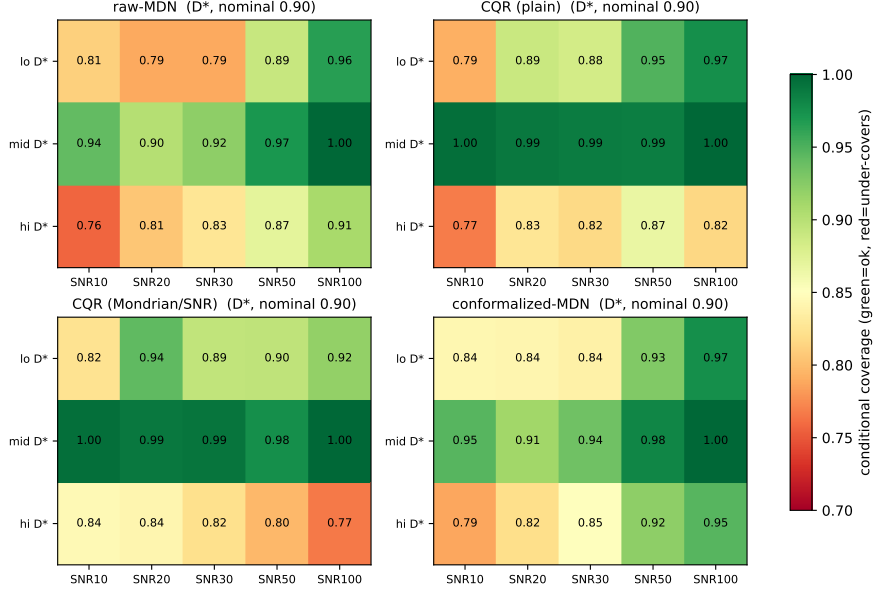


Figure 3: SNR $\times$ D^* -regime conditional coverage. The high- D^* row under-covers robustly at low-to-moderate SNR (recovering toward nominal only at the highest SNR) and resists SNR-Mondrian.

misspecification: a shift in the conditional law $P(y | x)$ is, in general, not repairable by an x -space likelihood ratio (weighting over-corrects D^* to 0.933 [0.924, 0.941], leaving a per-parameter spread of 0.033).

The label-free monitor (reusing the deployment-validity concept of a sibling project) **fires on all four observable breaks before coverage fails**: in an SNR-severity sweep, it raises the alarm at SNR 20 (coverage still 0.866) while coverage first crosses the failure line at SNR 13. Yet it is constitutively *blind* to the latent high- D^* gap: on an in-distribution (exchangeable) test the monitor is silent (AUC 0.50) and marginal D^* coverage is nominal (0.898 [0.880, 0.913]), while the high- D^* conditional coverage is 0.814 [0.787, 0.839]. Observable shifts are caught; the within-distribution latent-axis failure is not – the same observable-vs-hidden split that recurs across this line of work.

3.5 Acquisition sensitivity: the wall is b-scheme-robust

Does the high- D^* wall move with the b-value scheme? We compare a clinical (11-value), a CRLB-optimal (11-value, greedy D^* -CRLB design), and a dense (22-value) scheme at fixed prior/SNR/seed (Table 4, Fig. 7). The high- D^* marginal coverage barely moves (0.821 $\rightarrow$ 0.835 $\rightarrow$ 0.808). The resolution ratio $\text{CRLB}(D^*)/\text{tercile-width}$ stays at or above unity: even a CRLB-optimal design only reaches $\sim$ unity (1.03 [0.97, 1.16]); acquisition shifts the wall to threshold, not past it, while the clinical and dense schemes stay above it (1.24, 1.11). A CRLB-optimal design lowers the ratio (1.24 $\rightarrow$ 1.03) but does not remove the wall. Acquisition design *shifts* the wall; it does not erase it – a concrete, honestly negative handoff to acquisition-aware experiment design.

3.6 In-vivo: a qualitative demonstration, no coverage claim

In vivo there are no ground-truth parameters, so realized coverage is *unmeasurable* and the finite-sample guarantee is *not asserted*. We demonstrate the deployed pipeline qualitatively (Fig. 8): the

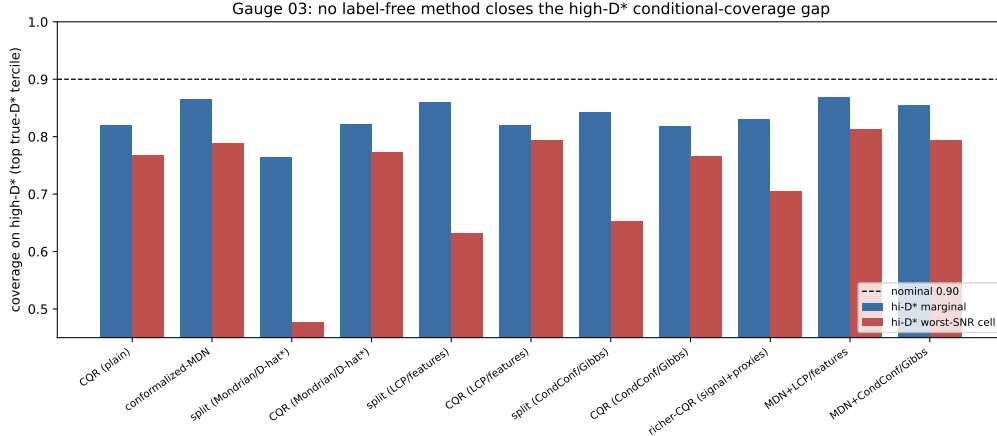


Figure 4: High- D^* coverage (marginal and worst-SNR cell) for eleven label-free conditioning methods; every bar sits below the nominal 0.90.

Table 3: Per-parameter coverage under each break, naive vs. weighted conformal ($\alpha = 0.10$, nominal 0.90), with the deployment monitor’s verdict. Across-seed means (16 seeds); per-entry [5, 95] bands are in the supplementary CI table (`results/ci_for_latex.json`). Monitor AUCs are the deterministic seed-0 values. From `results/multiseed_report.txt`.

scenario	method	D	D^*	f	monitor
in-dist (control)	naive	0.898	0.898	0.901	silent (AUC 0.50)
SNR shift (low)	naive	0.592	0.739	0.614	FIRES (AUC 0.91)
	weighted	0.882	0.901	0.909	
prior shift (harder)	naive	0.972	0.508	0.909	FIRES (AUC 0.71)
	weighted	0.919	0.991	0.902	
tri-exp misspec	naive	0.907	0.843	0.912	FIRES (AUC 0.51)
	weighted	0.891	0.933	0.898	
noise-model swap	naive	0.912	0.893	0.906	FIRES (AUC 0.52)

conformalized D^* band widths are wide and heavy-tailed (10/50/90th percentile = $47/64/79 \times 10^{-3} \text{ mm}^2/\text{s}$; widest-to-narrowest decile ratio $1.7\times$), ballooning on voxels where the perfusion compartment is under-identified – the wall, now on signal. Pointed at the deployment data with a synthetic calibration, the deployment monitor **fires** (AUC 0.84): synthetic→deployment is itself a large exchangeability break, and the monitor detects it. That firing is the honest, label-free reason the synthetic guarantee must not be transferred in vivo without re-calibration on matched labeled data – which in vivo does not exist. The monitor turns “no ground truth” into an observable stop sign. (No usable permissively-licensed, ground-truth-free in-vivo IVIM dataset was available for this study; the demonstration runs on a clearly labeled synthetic stand-in, and dataset choice/licensing and the in-vivo framing are deferred to human sign-off.)

4 Discussion

The thread connecting these results is an *observable-vs-latent* distinction. Conformal prediction guarantees *marginal* coverage and, with weighting or stratification on *observable* quantities (SNR, signal features, estimated density ratios), can recover coverage conditional on those observables.

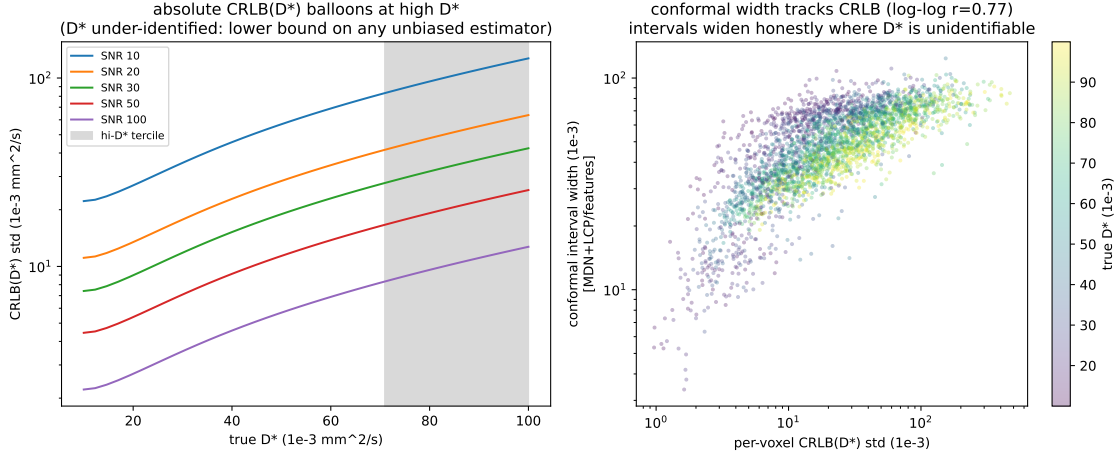


Figure 5: (Left) absolute $\text{CRLB}(D^*)$ balloons across the D^* range per SNR (high- D^* tercile shaded). (Right) conformal width vs per-voxel CRLB ($r = 0.76$): the intervals widen honestly where D^* is under-identified.

Table 4: High- D^* coverage and resolution ratio by b-value scheme. Across-seed means with 16-seed [5, 95] bands; the CRLB-optimal ratio band crosses unity. From `results/multiseed_report.txt`.

scheme	n_b	hi- D^* marg	hi- D^* worst-SNR	CRLB/tercile-width
clinical	11	0.821 [0.794, 0.845]	0.596 [0.542, 0.702]	1.24 [1.17, 1.37]
CRLB-optimal	11	0.835 [0.795, 0.862]	0.603 [0.541, 0.678]	1.03 [0.97, 1.16]
dense	22	0.808 [0.788, 0.836]	0.551 [0.508, 0.597]	1.11 [1.05, 1.21]

What it cannot label-free-guarantee is coverage conditional on a *latent* axis that the data do not identify. In IVIM that latent axis is the true D^* in its ill-posed regime: the Fisher information there collapses, the CRLB exceeds the bin width, and no observable proxy – including the plug-in $\hat{D}^*$ – recovers the regime. This is why an observable deployment monitor catches every *distribution* shift but is blind to the *within-distribution* high- D^* gap.

The same wall appears in adjacent label-free problems: deployment-validity monitoring is provably blind to shifts that leave observable statistics unchanged (a “hidden channel”), and the absence of in-vivo ground truth (the “Echo tension”) forecloses any direct coverage check on real data. We read these not as failures of conformal prediction – whose guarantee is exactly as advertised – but as a recurring *identifiability* boundary: distribution-free tools deliver everything the data identify and nothing they do not. Practically, this argues for (i) conformalizing a strong model-based base (sharpest valid intervals), (ii) deploying an observable monitor to guard exchangeability, and (iii) reporting the high- D^* compartment as characterized-but-unresolved rather than trusting a point map there.

5 Limitations

Our coverage results are *synthetic-primary* by necessity: conformal calibration needs labels that in-vivo IVIM lacks, so the guarantee is established on a forward model we control, and the in-vivo component is explicitly qualitative with no coverage claim. The synthetic cohort uses uniform priors over published ranges and Rician noise; real tissue is more heterogeneous, and the tri-exponential

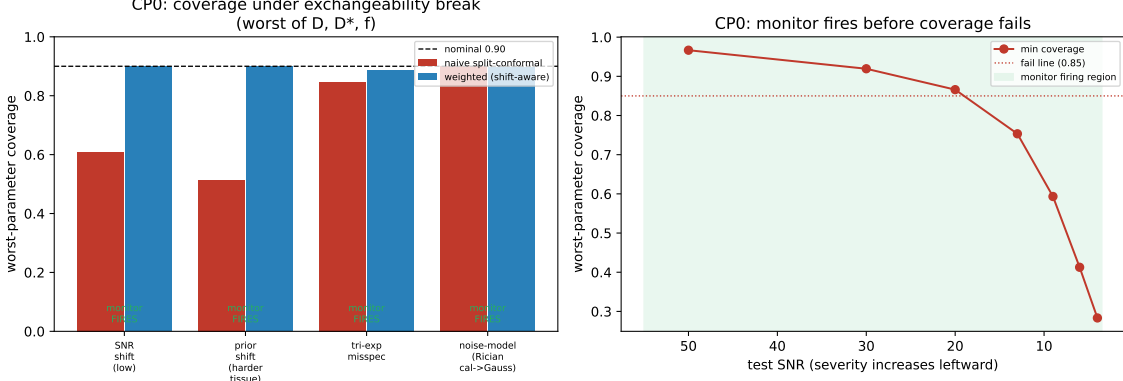


Figure 6: (Left) worst-parameter coverage under each exchangeability break, naive vs. weighted, with monitor status. (Right) SNR-severity sweep: the monitor fires before coverage crosses the failure line.

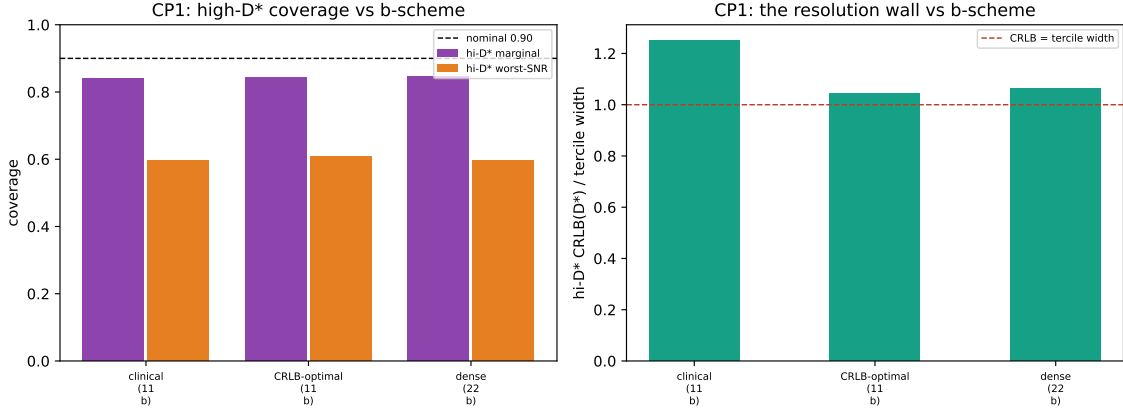


Figure 7: (Left) high- D^* coverage vs. b-scheme. (Right) the resolution ratio $\text{CRLB}(D^*)/\text{tercile-width}$ stays at or above unity for every scheme (the CRLB-optimal band dips to 0.97): acquisition moves the regime to the resolution threshold, not past it.

and prior-shift stresses (Sec. 3.4) are deliberately schematic probes of misspecification rather than a tissue model. The CRLB diagnostic uses the high-SNR Gaussian approximation to Rician noise; the qualitative under-identification of D^* is noise-model-independent (the D^* Jacobian column vanishes as D^* grows), but absolute bound values at $\text{SNR} \lesssim 10$ would shift under an exact Rician treatment. Finally, all coverage guarantees assume exchangeability of calibration and deployment data; the monitor quantifies, but cannot by itself repair, departures from it.

6 Consistency check (reproducibility)

Every numeric claim in this manuscript is drawn from a gated, seeded checkpoint printout. The script `gauge/paper/consistency.py` enforces two contracts: (i) every finite-sample-guaranteed marginal value and every determinism-gated quantity appears *verbatim* in its seed-20260613 report (`results/*.txt`, `POSITIONING.md`); and (ii) every conditional / empirical (E) headline ships as the *across-seed mean* over a 16-seed re-run and is asserted to lie within its empirical [5,95]

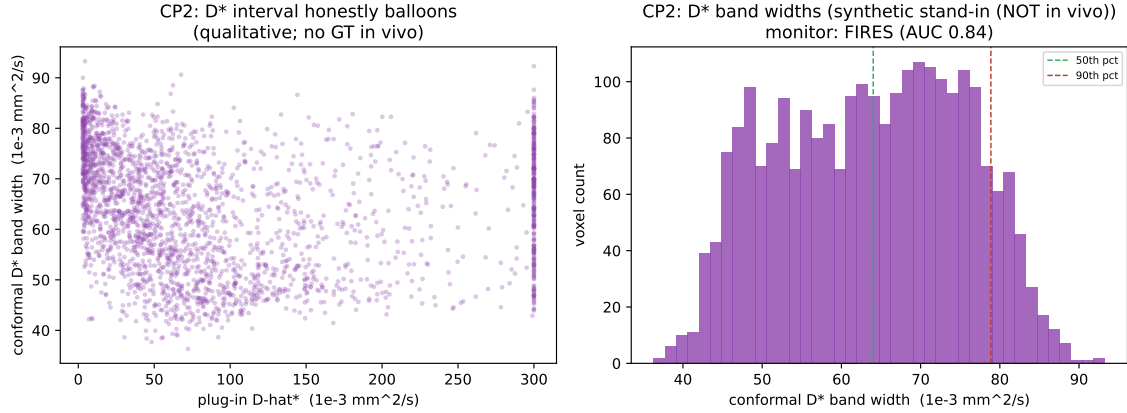


Figure 8: In-vivo qualitative demo (synthetic stand-in). (Left) the conformal D^* band balloons with the plug-in $\hat{D}^*$. (Right) the heavy-tailed D^* band-width distribution; the deployment monitor fires on the transfer.

band, which carries an `n_seeds` (`results/multiseed.json`, `multiseed_report.txt`; old→new point map in `results/ci_rebaseline_table.txt`). The full pipeline regenerates deterministically from seed 20260613 (`python -m gauge.baselines; gauge.benchmark; gauge.conditional; gauge.conditional_attack; gauge.robustness; gauge.invivo`), and the bands from the multi-seed driver (`python -m gauge.multiseed`); seed-0 outputs are byte-identical across independent full re-runs.

7 Conclusion

Gauge brings distribution-free conformal coverage to the ill-posed IVIM inverse problem and benchmarks it against model-based UQ. The naive hypothesis – that the conformal advantage concentrates marginally in D^*/f – is rejected: model-based IVIM UQ is overconfident broadly, conformal restores coverage everywhere, and conformalizing the MDN is the sharpest valid recipe. The genuine IVIM-specific phenomenon is conditional: the high- D^* regime under-covers universally, and a Fisher-information analysis shows this is an irreducible identifiability limit that no label-free method can close. We therefore characterize the limit – and provide the robustness tooling, acquisition analysis, and qualitative in-vivo demonstration that frame how an honest, distribution-free IVIM uncertainty map should be deployed and monitored.

Data and Code Availability Statement

This study uses no human or animal data. All results are produced from a deterministic, seeded synthetic cohort (seed 20260613) generated by the study’s own forward model; no external or in-vivo dataset is required to reproduce them. The complete source code, the gated checkpoint outputs from which every number in this paper is drawn, the figures, and a machine-checked consistency script (`gauge/paper/consistency.py`) are openly available in the project repository and permanently archived at <https://doi.org/10.5281/zenodo.20686274>.

Conflict of Interest

The author declares no competing interests.

Funding

This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors; it was conducted independently by the author.

Author Contributions

Avery Karlin is the sole author and conceived the study, designed and implemented the methods and software, performed the analysis, produced the figures, and wrote the manuscript.

ORCID

Avery Karlin <https://orcid.org/0000-0003-3848-6782>

References

- [1] Denis Le Bihan, Eric Breton, Denis Lallemand, M-L Aubin, J Vignaud, and M Laval-Jeantet. Separation of diffusion and perfusion in intravoxel incoherent motion MR imaging. *Radiology*, 168(2):497–505, 1988.
- [2] et al. Casali. A comprehensive framework for uncertainty quantification of voxel-wise supervised models in IVIM MRI. *NMR in Biomedicine*, 2026. arXiv:2508.04588.
- [3] Vladimir Vovk, Alexander Gammerman, and Glenn Shafer. *Algorithmic Learning in a Random World*. Springer, 2005.
- [4] Jing Lei, Max G'Sell, Alessandro Rinaldo, Ryan J Tibshirani, and Larry Wasserman. Distribution-free predictive inference for regression. *Journal of the American Statistical Association*, 113(523):1094–1111, 2018.
- [5] Anastasios N Angelopoulos and Stephen Bates. Conformal prediction: A gentle introduction. *Foundations and Trends in Machine Learning*, 16(4):494–591, 2023.
- [6] Yaniv Romano, Evan Patterson, and Emmanuel J Candès. Conformalized quantile regression. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2019. arXiv:1905.03222.
- [7] Felix Birk, Lucas Mahler, Julius Steiglechner, Qi Wang, Klaus Scheffler, and Rahel Heule. Distribution-free uncertainty estimation in multi-parametric quantitative MRI through conformalized quantile regression. Proc. Intl. Soc. Mag. Reson. Med. (ISMRM), abstract #2228, 2024. <https://archive.ismr.org/2024/2228.html>.
- [8] Christopher M Bishop. Mixture density networks. *Technical Report NCRG/94/004*, Aston University, 1994.

- [9] Balaji Lakshminarayanan, Alexander Pritzel, and Charles Blundell. Simple and scalable predictive uncertainty estimation using deep ensembles. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.
- [10] Harald Cramér. Mathematical methods of statistics. *Princeton University Press*, 1946.
- [11] Ryan J Tibshirani, Rina Foygel Barber, Emmanuel J Candès, and Aaditya Ramdas. Conformal prediction under covariate shift. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2019. arXiv:1904.06019.
- [12] Rina Foygel Barber, Emmanuel J Candès, Aaditya Ramdas, and Ryan J Tibshirani. Conformal prediction beyond exchangeability. *The Annals of Statistics*, 51(2):816–845, 2023.
- [13] Laying Guan. Localized conformal prediction: a generalized inference framework for conformal prediction. *Biometrika*, 110(1):33–50, 2023.
- [14] Isaac Gibbs, John J Cherian, and Emmanuel J Candès. Conformal prediction with conditional guarantees. *arXiv preprint arXiv:2305.12616*, 2023.